Simultaneous Cubic Walsh Modes in the Four-Bit Plain Heart Inequality

A complete all-channel proof for the pair-free, top-free odd sector

Research note

23 July 2026

Abstract

We prove the equal-channel plain Heart inequality for the genuinely four-coordinate family

$$f_d = 1 + \sum_{i=1}^{4} d_i \chi_{[4]\setminus\{i\}}, \qquad \sum_i |d_i| \le 1,$$

for every channel strength $0 \le s \le 1$. The four degree-three Walsh coefficients may be present simultaneously and with arbitrary signs. The proof converts the classical reveal entropy and the quantum relative entropy into exact power series indexed by closed words in four commuting translations of $\mathbb{F}_2^3$. A Hölder reduction bounds every quantum word by a universal pure-word coefficient. The first coefficient is controlled directly, the three fourth-order mixed word types are handled by explicit rational inequalities, and all higher orders follow from a monotonicity theorem for beta-integral moments. The inequality is strict for $0 < s < 1$ unless $d = 0$.

Contents

1 Statement and notation **2**

2 The exact classical series **3**

3 The exact quantum log-determinant series **4**

4 Closed words and the Hölder reduction **5**

5 Monotonicity of the pure-word moments **6**

6 The first coefficient **7**

7 The fourth-order coefficient **7**

8 All higher coefficients **9**

9 Completion and equality **10**

10 Discussion **10**

A The parity-block coordinate table **10**

1 Statement and notation

Let U_4 be the uniform law on $\{\pm 1\}^4$. For $S \subseteq [4]$, write

$$\chi_S(x) = \prod_{i \in S} x_i.$$

Fix $0 \le s \le 1$, put

$$\mu = \sqrt{1 - s^2}, \qquad |\psi_x\rangle = \sqrt{\frac{1+\mu}{2}}\,|0\rangle + x\sqrt{\frac{1-\mu}{2}}\,|1\rangle, \qquad x \in \{\pm 1\},$$

and define the product pure-state channel

$$\Phi_s(P) = \sum_{x \in \{\pm 1\}^4} P(x) \bigotimes_{j=1}^{4} |\psi_{x_j}\rangle\langle\psi_{x_j}|.$$

The uniform input produces

$$\sigma_4 = \sigma^{\otimes 4}, \qquad \sigma = \operatorname{diag}\left(\frac{1+\mu}{2}, \frac{1-\mu}{2}\right).$$

Independently reveal each input coordinate with probability s, let $R \subseteq [4]$ be the random revealed set, and write P_R for the corresponding marginal. For a density f with respect to U_4, put $P = fU_4$ and define

$$E_s(f) = \mathbb{E}_R D(P_R \| U_R), \qquad Q_s(f) = D(\Phi_s(P) \| \sigma_4), \qquad G_s(f) = E_s(f) - Q_s(f).$$

All logarithms are natural, with $0 \log 0 = 0$.

For $i = 1, \dots, 4$, set

$$T_i = [4] \setminus \{i\},$$

and consider

$$f_d = 1 + \sum_{i=1}^{4} d_i \chi_{T_i}. \tag{1}$$

The map

$$x \longmapsto \big(\chi_{T_1}(x), \dots, \chi_{T_4}(x)\big)$$

is a bijection from $\{\pm 1\}^4$ to $\{\pm 1\}^4$: if $y_i = \chi_{T_i}(x)$, then

$$x_i = \frac{\prod_j y_j}{y_i}.$$

Therefore the sixteen values of f_d are precisely $1 + \sum_i d_i y_i$, $y \in \{\pm 1\}^4$, and hence

$$f_d \ge 0 \quad \Longleftrightarrow \quad \sum_{i=1}^{4} |d_i| \le 1. \tag{2}$$

Theorem 1.1 (Simultaneous degree-three modes)**.** *For every $d \in \mathbb{R}^4$ satisfying equation* (2) *and every $0 \le s \le 1$,*

$$G_s(f_d) \ge 0.$$

If $d \ne 0$ and $0 < s < 1$, then $G_s(f_d) > 0$.

Proof architecture. The classical and quantum quantities admit exact expansions

$$G_s(f_d) = \frac{s^4}{16} \sum_{m \geq 1} \frac{\Gamma_m(d, s)}{m(2m-1)}.$$

The coefficient Γ_m is a sum over closed words of length $2m$ in four commuting translations. Every quantum word is bounded by a universal pure-word integral A_m. We prove the required bounds separately for $m = 1$, $m = 2$, and $m \geq 3$.

2 The exact classical series

Define the binary entropy increment

$$h(r) = \frac{(1+r)\log(1+r) + (1-r)\log(1-r)}{2} = \sum_{m \geq 1} \frac{r^{2m}}{2m(2m-1)}. \tag{3}$$

The series is valid for $|r| \leq 1$, with the endpoint values understood by continuity.

Every reveal of at most two coordinates annihilates all four characters χ_{T_i}. Revealing exactly T_i gives the density $1 + d_i \chi_{T_i}$, whereas revealing all four coordinates gives f_d. Therefore

$$E_s(f_d) = s^3(1-s) \sum_{i=1}^{4} h(d_i) + s^4 \mathbb{E}[f_d \log f_d]. \tag{4}$$

We encode the full-reveal term on an eight-point quotient. Let

$$\mathsf{G} = \mathbb{F}_2^3, \qquad g_1 = 6, \quad g_2 = 5, \quad g_3 = 3, \quad g_4 = 7,$$

where the integers are read as three-bit vectors, and let P_i be translation by g_i on $\ell_2(\mathsf{G})$. Thus each P_i is a symmetric involutive permutation matrix, and the P_i commute. Put

$$K = \sum_{i=1}^{4} d_i P_i. \tag{5}$$

The Walsh operator of f_d decomposes as

$$A_{f_d} = \begin{pmatrix} I & K \\ K & I \end{pmatrix} \tag{6}$$

under the even/odd parity decomposition. Hence, if k_t ($t \in \mathsf{G}$) are the eigenvalues of K, the eigenvalues of A_{f_d}—equivalently, the values of f_d—are $1 \pm k_t$. It follows that

$$\mathbb{E}[f_d \log f_d] = \frac{1}{8} \sum_{t \in \mathsf{G}} h(k_t). \tag{7}$$

Since $\operatorname{Tr} K^{2m} = \sum_t k_t^{2m}$, substituting equations (3) and (7) into equation (4) gives the exact formula

$$\boxed{E_s(f_d) = \frac{s^4}{16} \sum_{m \geq 1} \frac{\operatorname{Tr} K^{2m} + 8\frac{1-s}{s} \sum_i d_i^{2m}}{m(2m-1)}.} \tag{8}$$

For $s = 0$, this is interpreted by continuity.

3 The exact quantum log-determinant series

Assume first that $0 < s < 1$ and write

$$s = \operatorname{sech}\tau, \qquad R = e^{2\tau} > 1. \tag{9}$$

Index even and odd subsets of $[4]$ by $t = (t_1, t_2, t_3) \in \mathsf{G}$ via

$$E_t = (t_1, t_2, t_3, t_1 + t_2 + t_3),$$

$$O_t = (t_1, t_2, t_3, 1 + t_1 + t_2 + t_3).$$

Define

$$h_E(t) = 2 - |E_t|, \qquad h_O(t) = 2 - |O_t|.$$

In the order $t = 0, 1, \ldots, 7$,

$$h_E = (2, 0, 0, 0, 0, 0, 0, -2), \qquad h_O = (1, 1, 1, -1, 1, -1, -1, -1). \tag{10}$$

Let

$$(D_E^2)_{tt} = R^{h_E(t)}, \qquad (D_O^2)_{tt} = R^{h_O(t)}.$$

After extracting the common scalar $(s/2)^4$, the channel output corresponding to equation (1) is

$$\mathcal{R}_K = \begin{pmatrix} D_E^2 & D_E K D_O \\ D_O K D_E & D_O^2 \end{pmatrix}, \tag{11}$$

and the uniform output corresponds to $K = 0$.

The uniform-marginal condition implies

$$Q_s(f_d) = \operatorname{Tr}(\rho_d \log \rho_d) - \operatorname{Tr}(\sigma_4 \log \sigma_4),$$

because $\rho_d = \Phi_s(f_d U_4)$ and σ_4 have the same one-coordinate marginals. The scalar factor $(s/2)^4$ cancels in this equal-trace difference. We use the elementary identity

Lemma 3.1 (Equal-trace logarithmic determinant formula)**.** *If $X, Y \succeq 0$ have the same dimension and trace, then*

$$\operatorname{Tr}(X \log X) - \operatorname{Tr}(Y \log Y) = -\int_0^\infty \log \frac{\det(X + vI)}{\det(Y + vI)}\, dv.$$

Proof. For positive scalars, the second derivative in x of the right-hand side is $1/x$, the second derivative of $x \log x$. The two scalar expressions therefore differ by an affine function. Summing over eigenvalues, the constant terms cancel because the dimensions agree and the linear terms cancel because the traces agree. Positive semidefinite matrices follow by adding εI and letting $\varepsilon \downarrow 0$. □

For $v > 0$, set

$$U_v = \frac{D_E^2}{D_E^2 + vI}, \qquad V_v = \frac{D_O^2}{D_O^2 + vI}. \tag{12}$$

A Schur-complement calculation gives

$$\frac{\det(\mathcal{R}_K + vI)}{\det(\mathcal{R}_0 + vI)} = \det(I - V_v K U_v K). \tag{13}$$

Indeed, after factoring $D_E^2 + vI$ and $D_O^2 + vI$, the nontrivial Schur complement is

$$I - (D_O^2 + vI)^{-1} D_O K D_E (D_E^2 + vI)^{-1} D_E K D_O,$$

which is similar to $I - V_v K U_v K$.

Applying theorem 3.1 and restoring the common factor yields

$$Q_s(f_d) = \frac{s^4}{16} \int_0^\infty -\log\det(I - V_v K U_v K)\, dv. \tag{14}$$

If $\sum_i |d_i| < 1$, then $\|K\| < 1$ and $0 \prec U_v, V_v \prec I$, so the logarithm may be expanded absolutely:

$$\boxed{Q_s(f_d) = \frac{s^4}{16} \sum_{m \geq 1} \frac{I_m(d,s)}{m},} \tag{15}$$

where

$$I_m(d,s) = \int_0^\infty \operatorname{Tr}(V_v K U_v K)^m\, dv. \tag{16}$$

The boundary $\sum_i |d_i| = 1$ follows by continuity.

Combining equations (8) and (15), we obtain

$$\boxed{G_s(f_d) = \frac{s^4}{16} \sum_{m \geq 1} \frac{\Gamma_m(d,s)}{m(2m-1)},} \tag{17}$$

where

$$\boxed{\Gamma_m = \operatorname{Tr} K^{2m} + 8\frac{1-s}{s} \sum_i d_i^{2m} - (2m-1) I_m.} \tag{18}$$

The proof of theorem 1.1 is therefore reduced to showing $\Gamma_m \geq 0$ for every $m \geq 1$.

4 Closed words and the Hölder reduction

Input sign flips preserve both Heart functionals. Their action on the four coefficients d_i is invertible over $\mathbb{F}_2$, so we may assume

$$d_i \geq 0 \qquad (i = 1, \ldots, 4). \tag{19}$$

For a word $w = (i_1, \ldots, i_{2m}) \in [4]^{2m}$, write

$$d_w = \prod_{r=1}^{2m} d_{i_r}.$$

Call w *closed* if

$$g_{i_1} \oplus \cdots \oplus g_{i_{2m}} = 0. \tag{20}$$

Since the product of the corresponding translations is translation by the xor in equation (20),

$$\operatorname{Tr} K^{2m} = 8 \sum_{w \text{ closed}} d_w. \tag{21}$$

Expanding equation (16) similarly gives

$$I_m = \sum_{w \text{ closed}} \mathcal{J}_w(R) d_w, \tag{22}$$

where $\mathcal{J}_w(R)$ is the integrated channel coefficient of w.

Fix v. Along the alternating even/odd path determined by a closed word, the coefficient has the form

$$\sum_{t\in\mathsf{G}}\prod_{r=1}^{m} a_r(t),$$

where each a_r is a translate of one of the functions

$$t \longmapsto V_v(t)U_v(t\oplus g_i).$$

Hölder's inequality gives

$$\sum_{t}\prod_{r=1}^{m} a_r(t) \le \prod_{r=1}^{m}\left(\sum_{t} a_r(t)^m\right)^{1/m}.$$

Translation invariance and the symmetry of the four generators imply that all factors on the right are equal. Hence

$$\boxed{\mathcal{J}_w(R) \le A_m(R),} \tag{23}$$

where the universal pure-word coefficient is

$$A_m(R) = \int_0^\infty \sum_{t\in\mathsf{G}} \big[V_v(t)U_v(t\oplus g_1)\big]^m\, dv. \tag{24}$$

The profile pairs $(h_E(t\oplus g_1), h_O(t))$ have multiplicities

profile pair	multiplicity
$(-2,1)$	1
$(-1,0)$	3
$(-1,2)$	1
$(0,1)$	3

Define

$$J_m(a,b) = \int_0^\infty \left(\frac{ab}{(v+a)(v+b)}\right)^m dv. \tag{25}$$

Then

$$\boxed{A_m = J_m(R^{-2},R) + 3J_m(R^{-1},1) + J_m(R^{-1},R^2) + 3J_m(1,R).} \tag{26}$$

5 Monotonicity of the pure-word moments

Lemma 5.1 (Beta-integral representation)**.** *For $a,b>0$ and $m\ge 1$,*

$$J_m(a,b) = ab\int_0^1 \frac{x^{m-3/2}}{\sqrt{4ab+(a-b)^2x}}\,dx. \tag{27}$$

Consequently, $(2m-1)J_m(a,b)$ is nonincreasing in m.

Proof. Set

$$x = \frac{ab}{(v+a)(v+b)}.$$

Solving the resulting quadratic equation for v and differentiating gives equation (27). If $X_m \sim \mathrm{Beta}(m - \frac{1}{2}, 1)$, then

$$(2m-1)J_m(a,b) = 2ab\,\mathbb{E}\left[\frac{1}{\sqrt{4ab + (a-b)^2 X_m}}\right].$$

The random variables X_m increase in stochastic order with m, whereas the displayed function decreases in its argument. This proves the monotonicity. □

Summing the four terms in equation (26) gives the key consequence

$$\boxed{(2m-1)A_m \text{ is nonincreasing in } m.} \tag{28}$$

6 The first coefficient

For $m = 1$,

$$J_1(a,b) = \frac{ab}{a-b}\log\frac{a}{b}$$

with the continuous diagonal interpretation. Substituting the four pairs in equation (26) gives

$$A_1(R) = \frac{3(R+1)^3 \log R}{(R-1)(R^2+R+1)}. \tag{29}$$

Since

$$s = \frac{2\sqrt{R}}{R+1},$$

and

$$\log R \le \frac{R-1}{\sqrt{R}}, \tag{30}$$

we obtain

$$A_1 \le \frac{3(R+1)^3}{\sqrt{R}(R^2+R+1)} \le \frac{4(R+1)}{\sqrt{R}} = \frac{8}{s}.$$

The second inequality is equivalent to

$$3(R+1)^2 \le 4(R^2+R+1),$$

whose defect is $(R-1)^2$.

A closed word of length two must be pure, so equations (18), (21) and (22) give

$$\Gamma_1 = \left(\frac{8}{s} - A_1\right)\sum_i d_i^2 \ge 0. \tag{31}$$

For $R > 1$, both inequalities above are strict; hence $\Gamma_1 > 0$ whenever $d \neq 0$ and $0 < s < 1$.

7 The fourth-order coefficient

For a pure word of length four, equations (28) and (29) imply

$$3A_2 \le A_1 \le \frac{8}{s}.$$

This is exactly the available classical coefficient: the full-reveal term contributes 8, and the three-coordinate reserve contributes $8(1-s)/s$.

Every mixed closed word of length four contains two copies of each of two distinct generators. Up to cyclic symmetry, the three patterns are

$$iijj, \qquad ijji, \qquad ijij.$$

Put

$$x = \tanh\tau \in [0,1), \qquad L = \operatorname{artanh} x. \tag{32}$$

A direct partial-fraction evaluation gives their integrated coefficients

$$W_A = \frac{L(x^8+6x^6-4x^4+18x^2-21)-x^7-x^5+13x^3+21x}{x^3(x^2+3)^2}, \tag{33}$$

$$W_B = \frac{x^9+6x^7+20x^5+34x^3+3x-L(x^{10}+11x^8+6x^6+18x^4-39x^2+3)}{x^3(x^2+3)^3}, \tag{34}$$

$$W_C = \frac{2(1-x^2)\left[L(x^4+12x^2+3)-x^3-3x\right]}{x^3(x^2+3)^2}. \tag{35}$$

Each has continuous value $8/3$ at $x = 0$.

We use the elementary truncation bounds

$$x + \frac{x^3}{3} \le L \le x + \frac{x^3}{3} + \frac{x^5}{5(1-x^2)}. \tag{36}$$

They follow immediately from the positive series

$$\operatorname{artanh} x = \sum_{j\ge 0} \frac{x^{2j+1}}{2j+1}.$$

Type A

After multiplying $8/3 - W_A$ by the positive denominator $3x^3(x^2+3)^2$, the numerator is

$$N_A = 3(1-x^2)(x^2+7)(x^4+3)L + 11x^7 + 51x^5 + 33x^3 - 63x.$$

Using the lower bound in equation (36),

$$N_A \ge -x^5(x^2+3)(x^4+6x^2-15) \ge 0. \tag{37}$$

Type B

Set

$$Q_B = x^8 + 12x^6 + 18x^4 + 36x^2 - 3.$$

The corresponding numerator is

$$N_B = 3(x^2-1)Q_B L + 5x^9 + 54x^7 + 156x^5 + 114x^3 - 9x.$$

If $Q_B \le 0$, the lower bound in equation (36) gives

$$N_B \ge x^5(x^2+3)(x^6+11x^4+11x^2+57) > 0. \tag{38}$$

If $Q_B \ge 0$, the upper bound gives

$$N_B \ge \frac{2x^5}{5}(x^8+17x^6+83x^4+171x^2+432) > 0. \tag{39}$$

Type C

Its numerator is

$$N_C = -3(1-x^2)(x^4+12x^2+3)L + 4x^7 + 21x^5 + 30x^3 + 9x.$$

The upper bound in equation (36) gives

$$N_C \geq \frac{2x^5}{5}(x^4+27x^2+108) > 0. \tag{40}$$

We have proved

$$\boxed{W_A, W_B, W_C \leq \frac{8}{3}.} \tag{41}$$

At $m = 2$, the quantum multiplier is $2m-1 = 3$, so every mixed word has quantum coefficient at most 8, exactly its coefficient in $\operatorname{Tr} K^4$. Together with the pure-word estimate, this proves

$$\Gamma_2 \geq 0. \tag{42}$$

8 All higher coefficients

A direct evaluation of equation (26) at $m = 3$ gives

$$A_3(x) = \frac{2L\,P(x) + Q(x)}{2x^5(x^2+3)^5}, \tag{43}$$

where

$$P(x) = 9(1-x^2)^2(5x^8+20x^6+70x^4+100x^2+61), \tag{44}$$

$$Q(x) = 2x(x^2+3)(43x^8+158x^6+300x^4+66x^2-183). \tag{45}$$

Use the sharper truncation

$$L \leq x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7(1-x^2)}. \tag{46}$$

Substitution and exact simplification give

$$\begin{aligned}
&\frac{16}{5}x^5(x^2+3)^5 - [2LP(x)+Q(x)] \\
&\quad \geq \frac{4x^7}{35}\left(3888 - 629x^2 - 340x^4 - 870x^6 - 212x^8 - 45x^{10}\right) \\
&\quad \geq \frac{4x^7}{35}\cdot 1792 \geq 0.
\end{aligned} \tag{47}$$

Thus

$$5A_3 \leq 8. \tag{48}$$

By equation (28),

$$(2m-1)A_m \leq 5A_3 \leq 8 \qquad (m \geq 3). \tag{49}$$

Combining equations (23) and (49), every closed word of length $2m$, $m \geq 3$, has quantum coefficient at most 8. This is bounded by its coefficient in $\operatorname{Tr} K^{2m}$; pure words also have the additional nonnegative three-coordinate reserve. Therefore

$$\Gamma_m \geq 0 \qquad (m \geq 3). \tag{50}$$

9 Completion and equality

The three estimates in equations (31), (42) and (50) show that every coefficient in the exact gap series equation (17) is nonnegative. Hence

$$G_s(f_d) \geq 0$$

for $0 < s < 1$ and $\sum_i |d_i| < 1$. The boundary $\sum_i |d_i| = 1$ follows by continuity, as do the channel endpoints $s = 0, 1$.

If $d \neq 0$ and $0 < s < 1$, the first coefficient equation (31) is strictly positive. Since every other coefficient is nonnegative, the entire gap is strict. This completes the proof of theorem 1.1.

10 Discussion

The theorem controls all four degree-three Walsh modes simultaneously. It is neither a tensor-product result nor a consequence of the known one-mode axes: a generic admissible vector d makes f_d depend essentially on all four input coordinates.

The proof also identifies a reusable mechanism for plain Heart:

1. pass to parity blocks and express the quantum relative entropy through a log-determinant integral;
2. expand in closed words of a finite translation group;
3. reduce arbitrary words to pure words by Hölder;
4. isolate finitely many low-order mixed patterns, then control the tail by a beta-integral monotonicity theorem.

The remaining four-bit odd problem is the interaction of these simultaneous cubic modes with a nonzero degree-four coefficient, or more generally with a nontrivial even base.

A The parity-block coordinate table

For completeness, the even and odd subset labels and channel exponents used in equation (10) are displayed below. Addition in the final coordinate is modulo two.

t	E_t	$h_E(t)$	O_t	$h_O(t)$
000	0000	2	0001	1
001	0011	0	0010	1
010	0101	0	0100	1
011	0110	0	0111	−1
100	1001	0	1000	1
101	1010	0	1011	−1
110	1100	0	1101	−1
111	1111	−2	1110	−1

For $g_1 = 110$, the multiset of pairs $(h_E(t \oplus g_1), h_O(t))$ is

$$(-2, 1), \quad 3 \times (-1, 0), \quad (-1, 2), \quad 3 \times (0, 1),$$

after translating exponents into the powers (R^{h_E}, R^{h_O}) appearing in equation (25). The other generators give the same multiset by coordinate symmetry.

B Elementary logarithmic bounds

The inequality equation (30) follows by setting $R = e^{2u}$:

$$\frac{R-1}{\sqrt{R}} = 2\sinh u \geq 2u = \log R.$$

The bounds equations (36) and (46) follow from the positive series

$$\operatorname{artanh} x = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \cdots$$

and the estimates

$$\sum_{j\geq 2} \frac{x^{2j+1}}{2j+1} \leq \frac{x^5}{5} \sum_{j\geq 0} x^{2j} = \frac{x^5}{5(1-x^2)},$$

$$\sum_{j\geq 3} \frac{x^{2j+1}}{2j+1} \leq \frac{x^7}{7} \sum_{j\geq 0} x^{2j} = \frac{x^7}{7(1-x^2)}.$$